

Fermi–Amaldi-Based Local Mixing Function

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I. MAIN TEXT

A hybrid exchange functional is called local when

$$E_x^{LH} = \int \{g(\mathbf{r})e_x^{\text{exact}}(\mathbf{r}) + [1 - g(\mathbf{r})]e_x^{\text{SL}}(\mathbf{r})\} d\mathbf{r}, \quad (1)$$

In the equation above, the integrand is a linear combination of two exchange densities multiplied by a position-dependent function, $g(\mathbf{r})$, called local mixing function (LMF). The e_x^{SL} and e_x^{exact} quantities are a semi-local and the exact exchange energy densities, respectively. The closed-shell form of the exact exchange density via one-electron reduced density matrix, $\gamma(\mathbf{r}; \mathbf{r}') = 2 \sum_{i=1}^{N/2} \varphi_i^*(\mathbf{r})\varphi_i(\mathbf{r}')$, is given by [1, 2]

$$e_x^{\text{exact}}(\mathbf{r}) = -\frac{1}{4} \int \frac{|\gamma(\mathbf{r}; \mathbf{r}')|^2}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r}'. \quad (2)$$

Here, we propose a new local mixing function for local hybrid functionals. This LMF is the ratio between the Fermi–Amaldi and the exact exchange energy densities:

$$g(\mathbf{r}) = \frac{e_x^{\text{FA}}}{e_x^{\text{exact}}}. \quad (3)$$

The Fermi–Amaldi [3, 4] energy density is expressed via electron density, $\rho(\mathbf{r}) \geq 0$, and electron number, N ,

$$e_x^{\text{FA}}(\mathbf{r}) = -\frac{1}{2N} \int \frac{\rho(\mathbf{r})\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r}'. \quad (4)$$

Both exchange densities are manifestly non-positive and equivalent for one-electron systems and two-electron singlets. Although, for $N > 2$, one can show that

$$e_x^{\text{exact}} \leq e_x^{\text{FA}} \leq 0, \quad (5)$$

since e_x^{FA} is scaled down by N . Combining these properties, we find that the proposed LMF is bound between zero and one:

$$0 \leq \frac{e_x^{\text{FA}}}{e_x^{\text{exact}}} \leq 1. \quad (6)$$

This inequality ensures that one hundred percent of exact exchange is employed for one-electron systems, satisfying the one-electron constraint [5, 6].

A particular case of a one-electron limit, i.e., the asymptotic limit ($r \rightarrow \infty$) where the exact exchange must cancel out an excessive self-repulsion of the Coulomb potential, is also satisfied:

$$\lim_{r \rightarrow \infty} \frac{e_x^{\text{FA}}}{e_x^{\text{exact}}} = 1 \quad (7)$$

since both energy densities approach $-1/2r$ as $r \rightarrow \infty$.

Another important constraint, the high-density limit,

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